7 Semantic segmentation

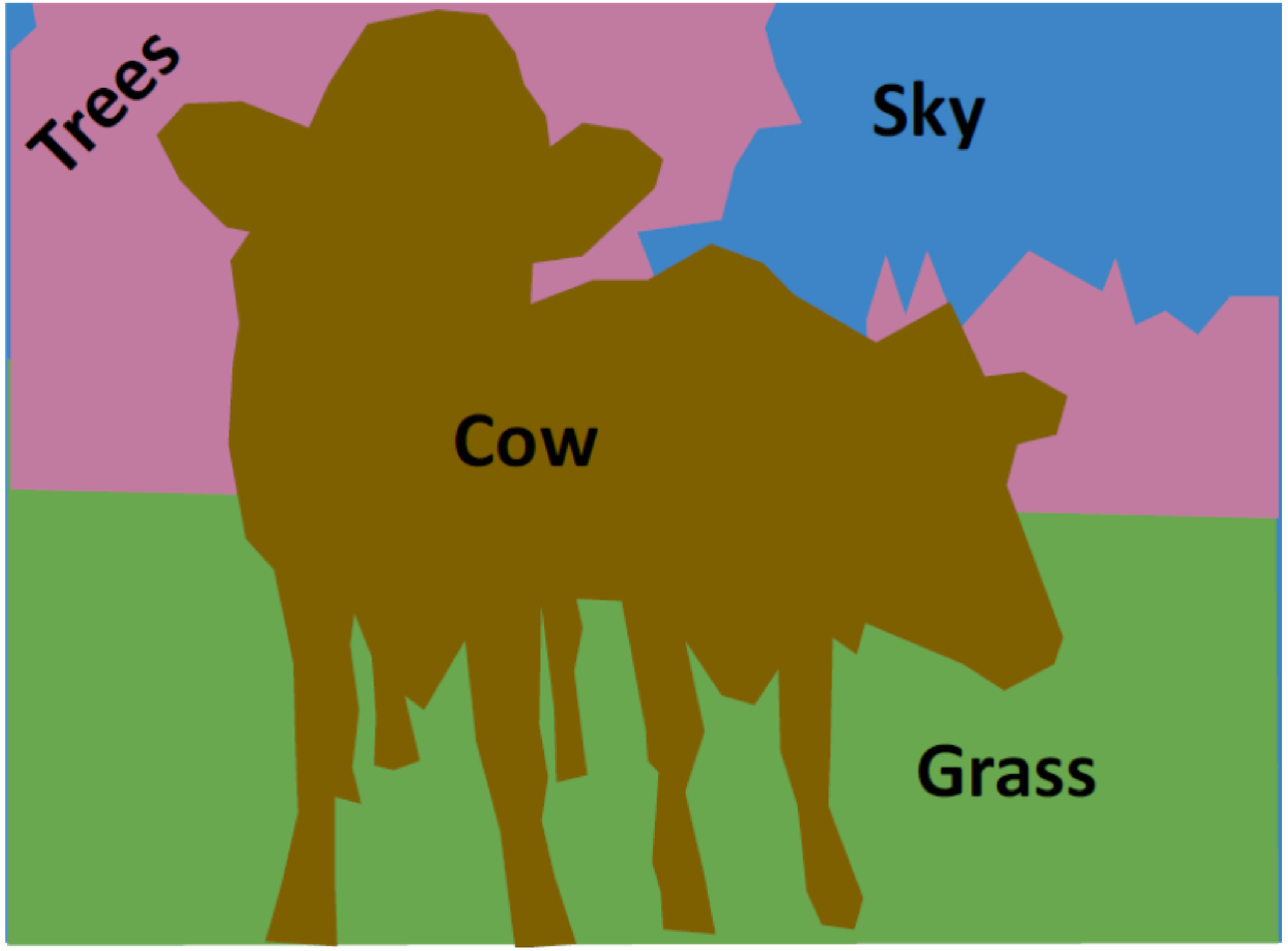
- when discussing object detection tasks in Chapter 6, rectangular bounding boxes are used to label and predict objects in images



7 Semantic segmentation

- this chapter will discuss the problem of *semantic segmentation*, which focuses on how to divide an image into regions belonging to different semantic classes
- unlike object detection, semantic segmentation recognizes and understands the content of images at pixel level: its labeling and prediction of semantic regions are at pixel level
- compared with the case of object detection, the pixel-level borders labeled in semantic segmentation are obviously more fine-grained

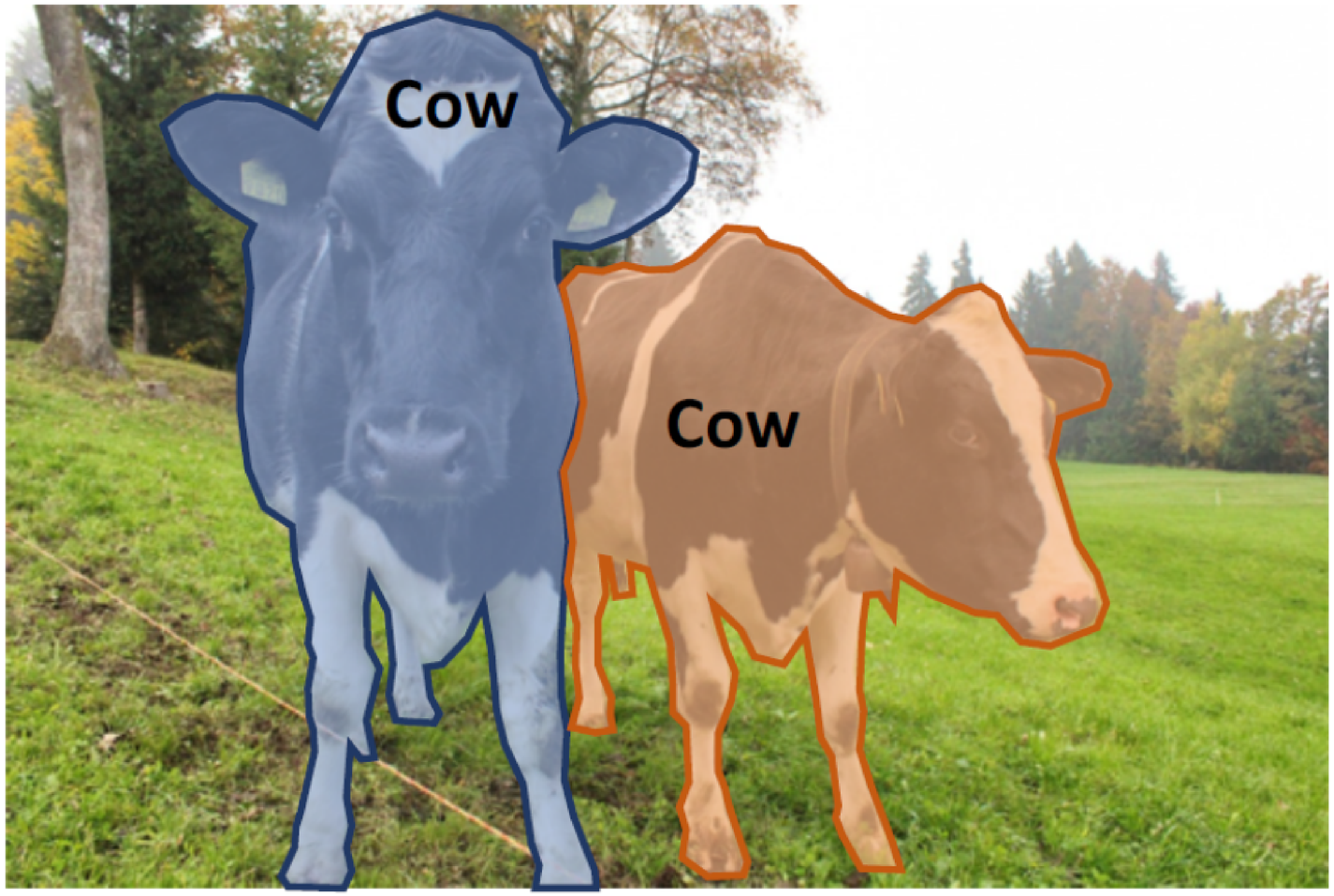
7 Semantic segmentation



7 Semantic segmentation

- there are also two important tasks in the field of computer vision that are similar to semantic segmentation, namely instance segmentation and panoptic segmentation
- *instance segmentation* is also called *simultaneous detection and segmentation*
- it studies how to recognize the pixel-level regions of each object instance in an image
- different from semantic segmentation, instance segmentation needs to distinguish not only semantics, but also different object instances
- for example, if there are two cows in the image, instance segmentation needs to distinguish which of the two cows a pixel belongs to

7 Semantic segmentation



7 Semantic segmentation

- *panoptic segmentation* gives labels to uncountable objects called "stuff" (sky, grass, etc.), similar to semantic segmentation, and differentiates between pixels coming from different instances of the same class (countable objects) called "things" (cows, etc.), similar to instance segmentation

